

ing a tractor, or a harvester, farmers in many states can now rent one for a short duration (depending on the size of their farm and their budget).

It's not just renting, there's also some innovation being done using driverless tech. While our roads may not be safe for driverless cars, our farms are certainly safe enough to have driverless tractors. Mahindra, for one, showcased their driverless tractor last year. In fact, they were supposed to be made available for purchase by farmers by now, but we've recently heard about semi-autonomous tractor being produced for select farmers, so it's probably still in testing and being perfected.

SALES AND SUPPLY

This is the field we've been perhaps the most successful in when it comes to technology being married to agriculture. India has an expertise in supply chain management, given that the complexities of transporting across our vast country, across state lines, are already high.

It's also the easiest to solve for, because we are adept at e-commerce solutions, and this is really no different than selling anything else. Setting up marketplaces for farmers, showing inventory, updating live prices (based on demand), and finding the best ways to send supplies across the country is something many startups in the agritech industry are doing. In fact, there are plenty of apps available to farmers, usually in their native language, to find buyers for their crops even before harvest season.

Artificial scarcities or flooding of a market by hoarders and middlemen can also be avoided using software and apps. If a hoarder is trying to drive down local prices, farmers can easily contact non-local buyers and negotiate for better deals on their crops. Even consumers benefit because wholesalers and retailers can connect directly with farmers or farming unions to buy things for cheaper and cut out the middlemen completely.

Technology can also be used to ensure that supply lines are being monitored, and that shortages just never happen.

FOOD STORAGE

The same supply chain management solutions are usually also used to do warehousing of foodgrains. While fresh vegetables and fruit are not possible to store traditionally, unless they're packaged and frozen. Grains and legumes are often stored for considerable times, and it's usually more important to store food closer to the destination than the source.

Since many startups are trying to use technology to completely eliminate the middlemen, by allowing farmers to directly sell to consumers, warehousing becomes super important. After all, no consumer wants to wait a week to get their groceries. Heck we Indians want to touch

off the plants ourselves. So the future of food storage warehouses might be that they will cease to exist!

CONCLUSION

While the broad categories we have listed here do cover most of the innovations happening in India in the agritech field, there are a few worthy mentions of technologies that are going to change the future of our country.

AI, obviously, is going to have a hand in all aspects of agritech. Solar power too, is going to be hot, especially in India, where sunlight is found in abundance in most states. Sensor technologies and sensor webs (IoT or not) will also play a big role in our agricultural future. Heck there



Mahindra and Mahindra's driverless tractor promises to let farmers focus more on farming and less on driving a tractor. Just control it with a tablet and the tractor does whatever you want!

and feel the groceries we buy, and then cook them the same day. This is where amazon style supply chain management is needed in India. It's not really happening on a grand scale, but imagine if you could order grain, fruit and vegetables, and it all showed up on your doorstep an hour later, already paid for, and coming straight from a nearby farm. Better yet, imagine using your backyard to grow something exotic, and then selling it for a premium to consumers living around you.

In fact, with an increasing interest in hydroponics and growing of food within city limits, many believe that the future will involve us all walking through hydroponic farms and picking our fruit and veggies straight

are even ideas for Uber-like services for farmers who own machinery. They drive their own machines and work other people's farms, and get paid for doing so, just like Ola or Uber. Instead of taking someone somewhere, you go to them. This can be used to supplement incomes of farmers who own tractors, or can even be taken up as a full time job by farmers who are fed up of gambling with the monsoon!

Remember to write to us with more ideas that you love from the agritech scene, and especially write in and tell us about your own innovations, if you're involved in agritech in India. Happy Independence Day; here's looking forward to a promising food future for our country. 🇮🇳